Accident Analysis and Prevention 37 (2005) 1135–1144

ACCIDENT ANALYSIS & PREVENTION

www.elsevier.com/locate/aap

The effects of speed enforcement with mobile radar on speed and accidents An evaluation study on rural roads in the Dutch province Friesland

Charles Goldenbeld*, Ingrid van Schagen

SWOV Institute for Road Safety Research, P.O. Box 1090, NL-2260 BB Leidschendam, Netherlands

Received 3 March 2005; received in revised form 13 June 2005; accepted 13 June 2005

Abstract

In an evaluation study, the effects of targeted speed enforcement on speed and road accidents were assessed. Enforcement was predominantly carried out by means of mobile radar and focused on rural non-motorway roads. Information and publicity supported the enforcement activities. The evaluation covered a period of 5 years of enforcement. The speed data of these 5 years and the year preceding the enforcement project showed a significant decrease in mean speed and the percentage speed limit violators over time. The largest decrease was found in the first year of the enforcement project and in the fourth year of the project, when the enforcement effort was further intensified. There were similar decreases in speeding at both the enforced roads and at the nearby comparison roads that were not subjected to the targeted speed enforcement project, which may be explained by spillover effects. The best estimate for the safety effect of the enforcement project is a reduction of 21% in both the number of injury accidents and the number of serious casualties. This was based on comparison between the number of accidents/casualties during the enforcement project (5 years) and and the 8 preceding years on the enforced roads and at all other roads outside urban areas in the same region.

Keywords: Police enforcement; Speed cameras; Evaluation; Speed; Road safety

1. Introduction

Excess and inappropriate speed is a very important factor in the road safety problem (ETSC, 1995). First of all, speed is related to the risk of getting involved in a road accident. Faster speeds leave less time to react to changes, they lead to longer stopping distances and to less maneuverability. Secondly, there is a direct relationship between impact speed and the severity of an accident (e.g. Nilsson, 1982; Elvik et al., 2004). Evans (2004) reported that a 1% increase in speed increases the fatality risk by 4–12%. With regard to speed and accident risk on rural roads, Kloeden et al. (2001) estimated that the risk of involvement in an injury accident is more than twice as high when traveling 10 km/h above the average speed of non-accident involved vehicles, and nearly six times higher when traveling 20 km/h above that average speed.

In the Netherlands, as in most other countries, exceeding the speed limit is a very common offence. On average 40–45% of the Dutch car drivers on a particular road exceed the posted speed limit (van Schagen et al., 2004). Police enforcement is one of the most commonly used instruments to reduce speed limit violations. In the Netherlands, in the late 1990s, speed enforcement got a new impetus with the launch of regional programs for intensified police traffic enforcement. These programs focused on five spearheads: speeding, drink-driving, red light running, seat belt use, and helmet use by moped riders. For each of the 25 police regions in the Netherlands, national, regional, and local

* Corresponding author. Tel.: +31 70 317 3333/3364;
fax: +31 70 320 1261.
E-mail address: charles.goldenbeld@swov.nl (C. Goldenbeld).

0001-4575/$ – see front matter © 2005 Elsevier Ltd. All rights reserved.
doi:10.1016/j.aap.2005.06.011

authorities developed a tailor-made enforcement plan, based on the analysis of the region's traffic safety problem. In each region, 28 policemen were added to the general constabulary to carry out enforcement tasks. Targets were set in terms of both effort indicators (i.e., the number of hours spent and fines issued) as well as in terms of effect indicators (i.e., a reduction in accidents and violators). The program was supported by regional publicity campaigns and financed by the expected revenues of the enforcement activities. The regional enforcement projects started off in eight police regions and, by the end of 2001, all 25 police regions were involved.

Preceding the full-scale regional enforcement program, a number of pilot projects were carried out. One of the pilot projects was conducted in the province of Friesland in the Northern part of the Netherlands. This pilot started in 1998 with a special interest in speeding on rural non-motorway roads. In January 2001 the project was upgraded to the full-scale program. The current study aimed to assess the effects of 5 years (1998–2002) of the speed enforcement activities at rural roads in the Province of Friesland on both speed behavior and road accidents.

Generally, reviews report positive effects of speed enforcement on speeding behavior and the number of accidents (ETSC, 1999; Pilkington and Kinra, 2005; Zaal, 1994; Zaidel, 2002). Recent evaluation studies report positive speed and safety effects for both conspicuous fixed and mobile cameras (Christie et al., 2003; Gains et al., 2004; Hess and Polak, 2003), and for hidden mobile cameras (Keall et al., 2001, 2002). The sizes of the reported effects of speed enforcement, however, vary largely. For example, Pilkington and Kinra (2005) found that the accident and Casualty reduction in the immediate vicinity of the speed camera sites, reported in evaluation studies, varied between 5 and 69% for accidents, 12–65% for injuries and 17–71% for fatalities. These differences most likely have to do with the type, intensity and location of the enforcement activities as well as the situation before the enforcement started. On the other hand there is a large consistency in the finding that speed enforcement effects are limited in terms of both time (e.g. Vaa, 1997) and space (e.g. Christie et al., 2003; Hess, 2004).

Given the fact that in many countries traffic law enforcement forms a central part of the road safety program, there is still a considerable limitation in the extent of the scientific knowledge about the most (cost-)efficient ways of enforcing traffic violations in general and speed violations in particular. One of the reasons, as pointed out by Elvik (2001), is that studies of police enforcement do not always describe the different aspects of the enforcement situation, such as the severity of penalties, the incidence of violations before the start of the enforcement, and the way enforcement actions were divided over place and time. This situation complicates the task of drawing lessons from evaluation studies. In addition, as noted by Zaidel (2002, p. 48): "Much of the evidence for the positive impact of increased police enforcement (as separate from new legislation) on safety comes from enforcement projects and experiments restricted to either selected roads, to few behaviors or to a limited period". With specific reference to camera enforcement of speeding, Pilkington and Kinra (2005) conclude that the level of evidence from evaluation studies is still poor.

In the light of foregoing considerations, there remains a need for further independent, well-documented evaluation studies of the effects of speed enforcement. The present study aims to contribute to the body of knowledge. It focuses on the speed and safety effects of mobile inconspicuous speed cameras on rural roads.

Table 1
The national legal and provincial characteristics of the enforcement project

Context	Element	Description
National		
	Fines for speed offence on non-motorways	€30 (up to 10 km/h) €45 (11–15 km/h) €55 (16–20 km/h) €90 (21–25 km/h) etc.
	Legislative context	Speed offences below 50 km over the limit fall within administrative law framework and the license plate holder receives a speeding ticket by post 2–3 weeks after detection
	License revocation for speeding	Only possible for speed offences at least 50 km/h in excess of the speed limit
Friesland		
	Area size	5741 km^2 of which 3388 km^2 ground surface and 2353 km^2 water
	Population size	630 000
	Size of police force	About 1200 officers
	Traffic health risk	314 seriously injured or fatal traffic accidents victims per 100 000 inhabitants (vs. 249 per 100 000 inhabitants in the Netherlands) in 2000–2001
	Number of license holders	Total of 305 000 motor vehicles of which 242 000 passenger cars (or 384 cars per 1000 population), 35 000 vans and 3270 trucks

2. Method

2.1. The enforcement project

The evaluated speed enforcement project took place in the province of Friesland, a fairly rural, sparsely populated province in the Northern part of the Netherlands. Legally, traffic enforcement followed the national framework. Fines start off with relatively low fines, increasing rapidly when the amount of speeding increases. Most speed offences fall within the administrative law. In case of automatic enforcement, the license plate holder rather than the driver is held responsible for the speed offence. The time between the detection of speed offence and reception of a speeding ticket was 2–3 weeks. Table 1 provides information about these provincial and legal characteristics.

The project started in January 1998. The enforcement activities were directed at rural 80 and 100 km/h single carriageway roads with a large number of police reported injury accidents in the period 1992–1996. A total of 28 road sections with a total length of 116 km were identified as having a high injury accident level and were subjected to the targeted enforcement. The speed enforcement was effectuated by mobile radar equipment from an inconspicuous car (using wet-film cameras). On average, each week there was 1–2 h of speed checks on each of the selected roads. On each of the enforced roads a special, posted road sign warned drivers that speed camera enforcement was possible. The warning sign was constantly present, independent of the actual presence of enforcement. During the hours of enforcement, an inconspicuous (police) car was parked alongside the road, generally out of sight. There was no message sign to inform

Table 2
The enforcement project

Element	Description
Start of project	January 1998
Method of speed control	In the period 1998–2000 speed enforcement took place with mobile radar from an inconspicuous police car; in later years additional instruments were used, but the mobile radar remained dominant.
Speed violation margins	Speed violation at 87 km/h when 80 km/h limit and 107 km/h when 100 km/h limit
Types of road selected for enforcement	Single carriageway rural roads with a speed limit of 80 or 100 km/h with above average absolute number of injury accidents in period 1992–1996
Time between violation and receiving ticket	On average 3 weeks
Number and length of road segments	28 above average dangerous road segments of which the 100 km/h road segments (5) had a total length of 28 km and the 80 km/h segments (23) of 88 km (respectively about 11 and 15% of the roads under supervision of the provincial road authority)
Communication along roadside	Special signs alongside the road informed car drivers that they were driving on a segment of road where speed enforcement could take place
General publicity	The project had its own name, logo and publicity officer. Almost weekly, the regional newspapers covered the results. A few times per year, local and regional television, radio, magazines etc. paid attention to the project
Costs of the intensified enforcement	The total material and salary costs of the project in the period 1998–2002 have been estimated at nearly 5 million euro, of which 130 000 euro were spent on publicity for the project

Table 3
Indicators of the speed enforcement efforts on rural roads per year

Speed check method	Output-indicator	Year			
		1998	1999	2000	2001
Radar checks from inconspicuous police car	Number of checked vehicles	1 641 531	1 249 469	1 152 107	2 852 600
	Number of hours speed check	5486	4066	3092	8029
	Number of offenders	95 428	72 153	58 713	83 838
Radar checks from hidden tripod outside police car	Number of checked vehicles	–	–	237 994	1 056 069
	Number of hours speed check	–	–	547	2400
	Number of offenders	–	–	18 340	60 787
Lasergun (with stopping of offender)	Number of checked vehicles	–	–	17 795	38 823
	Number of hours speed check	–	–	564	3326
	Number of offenders	–	–	813	2119
Other (surveillance, video car, lasercam)	Number of checked vehicles	–	–	5375	82 838
	Number of hours speed check	–	–	163	684
	Number of offenders	–	–	614	4543
Total	Number of checked vehicles	1 641 531	1 249 469	1 410 106	3 999 553
	Number of hours speed check	5486	4066	4366	14 439
	Number of offenders	95 428	72 153	77 954	148 064

the driver that he/she had actually been exposed to speed enforcement.

There was a lot of publicity and communication about the project. The project had its own name and logo and a special publicity officer worked for the project, ensuring that results of the project appeared almost weekly in regional newspapers. Occasionally, local radio and television paid attention to the project. The total costs of the project were around 1 million euro per year. Table 2 summarizes the most important characteristics of the enforcement project in Friesland.

Table 3 provides information on the actual speed enforcement activities on the rural roads over the years 1998–2001. Regrettably, no detailed figures were available for 2002, but according to authorities the level of enforcement has remained stable in 2002 compared to 2001. In the first 3 years of the project, between 4000 and 5500 operational hours of speed checks with mobile radar were conducted. In 2001, the fourth year of the project, the enforcement levels of the project increased to over 14,000 h and other enforcement instruments, such as laser gun, tripod, laser cam (laser gun in combination with digital camera), and video cars were added. The majority of the enforcement, however, continued to take place by mobile radar.

To be able to distribute the available effort effectively over the selected road sections, the enforcement activities and speeding levels were systematically monitored. Each 5–6 weeks, the project team, consisting of members of police, road authority, and justice department, decided on several operational issues for the next weeks. These issues typically included: (1) Speed data and the consequences for the enforcement operations. For example, if the level of offenders on particular enforced roads decreased to below 10%, effort would be transferred to other roads with less favorable results; (2) monitoring of police manpower spent on camera operations; (3) planning and contents of press releases and other media activities.

2.2. The evaluation study

2.2.1. Design

The current evaluation study was designed as a before-and-after study with an experimental (targeted speed enforcement) and a comparison (no targeted speed enforcement) condition. It focused on the effects on speed and on road safety. For the evaluation of the speed effects, the experimental group consisted of the 12 road sections with a speed limit of 80 km/h. For those 12 road sections complete speed data for the full period was available. The comparison group consisted of those 15 road sections for which speed data was available, and which were not assigned to the experimental group. The comparison roads also had a speed limit of 80 km/h and similar road design characteristics as the experimental roads. The average traffic flow, however, was substantially smaller on the comparison roads (3800 versus 7200 vehicles per 24 h in 1997). The before period was 1997, the first year that speed data was available, but no targeted speed enforcement took place. The after period was 1998–2002, the period during which speed enforcement activities took place.

Table 4
Design of the evaluation study

Evaluation of effects on speed[a]	
Experimental road sections	12 road sections (length 60 km) of enforced 80 km/h rural roads
Comparison road sections	15 road sections (length 51 km) of the non-enforced 80 km/h rural roads
Evaluation of effects on road accidents[b]	
Experimental road sections	28 rural road sections (length 116 km) of enforced 80 km/h (23) and 100 km/h (5)
Comparison road sections	All other non-enforced roads outside urban areas in the province of Friesland (length approximately 5200 km)

[a] Before period (1997), after period (1998–2002).
[b] Before period (1990–1997), after period (1998–2002).

For the evaluation of the road safety effect, the experimental group consisted of all 28-road sections that were selected for enforcement. The majority (23) of these sections had a speed limit of 80 km/h; the remaining five had a speed limit of 100 km/h roads. The comparison group consisted of all other roads outside urban areas in the province of Friesland. It was assumed that the comparison group represents a broad reference category representative of the general development of road safety of roads outside urban areas in Friesland. The before period was 1990–1997 and the after period 1998–2002. Table 4 provides and overview of the study design. It should be noted that the speed analysis did not include data of the 100 km/h roads. The number of these roads was too small to include them meaningfully in a separate statistical analysis. In the road safety analysis, the enforced 100 km/h roads were included because they could be analyzed in combination with the 80 km/h roads.

It is clear that the current study is not a full experimental study. It cannot be excluded that occasionally some speed enforcement occurred on roads in the comparison group. However, it can be assumed that the level of enforcement, if at all, has been low in comparison to the experimental roads. Another, much more important point is that it was not possible to assign the roads randomly to the enforcement and the comparison condition. The road authority selected the enforcement roads on the basis of their high number of injury accidents. Hence, statistical regression-to-the-mean is a possible source of bias in this study, which may lead to an overestimation of the effect of the enforcement intervention. In the discussion, we will come back to this issue.

2.2.2. Dependent and independent variables

The study evaluated the effects of speed enforcement. Hence, the main independent variable in this study was the absence or presence of speed enforcement.

To evaluate the effects on speed, both the mean speed of motor vehicles (all types) and the percentage of violators were analyzed. A violator was operationally defined as a driver who drove 87 km/h or faster. This coincides with the threshold for issuing a speeding ticket.

To evaluate the road safety effects, the number of serious traffic casualties (fatalities and in-patients) resulting from accidents in which at least one motor vehicle was involved, as well as the number of injury accidents (all severities) in which at least one motor vehicle was involved were analyzed.

2.2.3. *Speed and accident data*

Speed data were obtained from speed measurement induction loops. The speed of every passing vehicle was registered electronically per hour (24 h a day, 7 days a week). Every month, the data was downloaded from the roadside data box, checked on minimal quality criteria, and forwarded for further analyses.

For the present study, additional checks on possible errors in the speed measurement data were performed before the actual analyses took place. Per road section and per day, specially developed software checked first whether the 24 h traffic flow deviated over six times the standard deviation from the average 24 h traffic flow on that road section. If this was the case, the data for that particular day was marked as possible error. A further automatic check verified for each day and for each road section whether the speed data approached a normal distribution. For days with more than 2000 observations, the Kolmogorov–Smirnov one sample test was applied and for days with less than 2000 observations the Shapiro–Wilk test (both tests described in Stevens, 1996). If the speed data did not have a normal distribution, again the data for that particular day was marked as a possible error. In consultation with the supplier of the data, the province of Friesland, it was concluded that on most of these days the deviations were caused by a measurement error. These days were left out of the analysis. Some days with deviating traffic flow or speed distributions were kept in the analysis because the province of Friesland indicated that it was most likely the result of some special occurrence or event affecting traffic on that particular day on that road. All together, for less than 5% of the days, data was either missing or removed because of measurement errors.

The accident and casualty data was extracted from the national road accident database which contains all accidents registered by the police. With the help of a Geographical Information System (Planet GIS) and the available *x*–*y* coordinates, the accidents were assigned to the experimental (enforced) or the comparison (non-enforced) road sections.

2.2.4. *Data analysis*

Mean speed and the percentage speed violators were analyzed by an analysis of variance for repeated measures with time (T) as an independent within-subjects variable, the presence or absence of speed enforcement (E) as an independent between subjects variable, and the interaction $T \times E$ as a within-subjects effect. The annual averages of either mean speed or percentage of offenders were considered as the within-subjects repeated measures factor. The road sections were considered to be the subjects. The analysis tested overall change over this period and the interaction between measurement year and enforcement. It also tested the specific differences between subsequent measurement years, i.e.: tests of contrasts between 1997 versus 1998; 1998 versus 1999; 1999 versus 2000; 2000 versus 2001; 2001 versus 2002. Given the fact that the enforcement project started in January 1998, and that the level of enforcement was considerably intensified as from January 2001, the contrasts 1997 versus 1998 and 2000 versus 2001 are of special interest. For all analyses, a significance level of 95% was applied. To assess the effect on road safety the odds-ratios were calculated for both the number of serious casualties and the number of injury accidents.

Although there was no reason to assume that the autonomous trends in speed behavior and accidents were different for the enforcement roads and the comparison roads, time series analysis could have been a reasonable alternative. Based on the following considerations, it was decided not to do so:

1. Regarding the accident data, there were relatively few data points (13 years) and consequently, the model would be based on rather speculative assumptions. In theory, it is possible to disaggregate the accident data to monthly data. However, in that case the accident numbers are very small and would require advanced modeling techniques that are still under development (see e.g. Durbin and Koopman, 2000).
2. Regarding speed data there were even fewer data points (6 years). Whereas disaggregating to months is easier, preliminary analyses showed that it is was not possible to fit one model that adequately represented the time series of the more than 25 roads in the present study.

3. Results

3.1. *Effects on speed behavior*

Table 5 presents the results of the repeated measures analyses with enforcement as between subjects factor and the speed measures in subsequent years 1997–2002 as within-subjects repeated measures. These analyses were conducted for the mean speed and the percentage offenders.

The results show that the main effect for the within subjects factor time (1997–2002) was significant for both the mean speeds ($F(5, 125) = 10.3$; $p = 0.000$) and the percentage offenders ($F(5, 125) = 10.8$; $p = 0.000$). The effect size measures corresponding to these effects, the partial eta squared (η^2), were 0.29 and 0.30. Cohen (1988) characterizes $\eta^2 = 0.01$ as a small, $\eta^2 = 0.06$ as a medium, and $\eta^2 = 0.14$ as a large effect size. Mauchly's test of sphericity was signifi-

Table 5
Results of the repeated measures analyses with absence or presence of speed enforcement as a between subjects factor and measurement years (1997–2002) treated as within-subjects factor

Effects	Degrees of freedom, *F*-value, significance, and size of effect	
	Mean speed	Percentage of offenders
Within subjects effects		
Time (T)	$F(5, 125) = 10.2$; $p = 0.000$; $\eta^2 = 0.29$ *Huynh–Feldt corrected: $F(3, 144) = 10.2$; $p = 0.000$; $\eta^2 = 0.29$	$F(5, 125) = 10.8$; $p = 0.000$; $\eta^2 = 0.30$ *Huynh–Feldt corrected: $F(2, 60) = 10.8$; $p = 0.000$; $\eta^2 = 0.30$
$T \times E$	$F(5, 125) = 2.4$; $p = 0.038$; $\eta^2 = 0.09$ *Huynh–Feldt corrected: $F(3, 73) = 2.4$; $p = 0.072$; $\eta^2 = 0.09$	$F(5,125) = 1.9$; $p = 0.096$; $\eta^2 = 0.07$ *Huynh–Feldt corrected: $F(2, 60) = 1.9$; $p = 0.148$; $\eta^2 = 0.07$
Within subjects contrasts		
Time 1997 vs. 1998	$F(1, 25) = 18.6$; $p = 0.000$; $\eta^2 = 0.42$	$F(1, 25) = 28.3$; $p = 0.000$; $\eta^2 = 0.53$
$T \times E$ 1997 vs. 1998	$F(1, 25) = 1.3$; $p = 0.257$; $\eta^2 = 0.05$	$F(1, 25) = 1.5$; $p = 0.234$; $\eta^2 = 0.06$
Time 1998 vs. 1999	$F(1, 25) = 4.5$; $p = 0.044$; $\eta^2 = 0.15$	$F(1, 25) = 4.8$; $p = 0.038$; $\eta^2 = 0.16$
$T \times E$ 1998 vs. 1999	$F(1, 25) = 2.6$; $p = 0.116$; $\eta^2 = 0.10$	$F(1, 25) = 2.9$; $p = 0.099$; $\eta^2 = 0.10$
Time 1999 vs. 2000	$F(1, 25) = 8.5$; $p = 0.007$; $\eta^2 = 0.25$	$F(1, 25) = 7.6$; $p = 0.001$; $\eta^2 = 0.23$
$T \times E$ 1999 vs. 2000	$F(1, 25) = 1.0$; $p = 0.320$; $\eta^2 = 0.04$	$F(1, 25) = 1.2$; $p = 0.281$; $\eta^2 = 0.05$
Time 2000 vs. 2001	$F(1, 25) = 15.1$; $p = 0.001$; $\eta^2 = 0.38$	$F(1, 25) = 15.0$; $p = 0.001$; $\eta^2 = 0.37$
$T \times E$ 2000 vs. 2001	$F(1, 25) = 4.8$; $p = 0.037$; $\eta^2 = 0.16$	$F(1, 25) = 2.9$; $p = 0.100$; $\eta^2 = 0.10$
Time 2001 vs. 2002	$F(1, 25) = 8.4$; $p = 0.008$; $\eta^2 = 0.25$	$F(1, 25) = 8.8$; $p = 0.007$; $\eta^2 = 0.26$
$T \times E$ 2001 vs. 2002	$F(1, 25) = 1.7$; $p = 0.209$; $\eta^2 = 0.06$	$F(1, 25) = 1.1$; $p = 0.307$; $\eta^2 = 0.04$

cant for both of these analyses, indicating that the assumption of sphericity of the data (i.e., the assumption that all possible differences between within-subjects conditions have the same population variance) was not met. The Huynh–Feldt corrected repeated measures test, which takes account of this lack of sphericity, however, produced similar significant findings for the main effect of time. This means that there is an overall difference in the mean speed and the percentage offenders between the different measurement years. Figs. 1 and 2 shows that in fact the mean speeds and the percentage offenders decreased over time.

With regard to mean speed a significant interaction was found between time and enforcement ($F(5, 125) = 2.4$; $p = 0.038$; $\eta^2 = .09$). Again Mauchly's test of sphericity was significant indicating that a corrected test of within-subjects effects would be more appropriate. Using the Huynh–Feldt corrected repeated measures test, the interaction was not significant ($p = 0.72$), meaning that the mean speed decreased to the same extent at road sections subjected to targeted speed enforcement and at the comparison road sections. However, although not significant, Fig. 1 shows that there is a tendency that the decrease was larger on the enforced roads. From 1997 to 2002 the mean speed decreased on average with 4 km/h on the enforced roads and about 1.5 km/h on the comparison roads. With regard to the percentage offenders, the interaction between time and enforcement did not reach a significant level either ($p = 0.096$; Huynh–Feldt corrected $p = 0.148$), but again, as can be seen in Fig. 2, there is a tendency that the decrease was larger on enforced roads. The lack of significance is most likely to be attributed to insufficient discriminatory power of the statistical test due to large standard deviations (see Table 6).

The specific within-subjects contrasts show significant differences in the mean speed as well as the percentage offenders for all comparisons between a specific year and its preceding year (see Table 5). This means that from

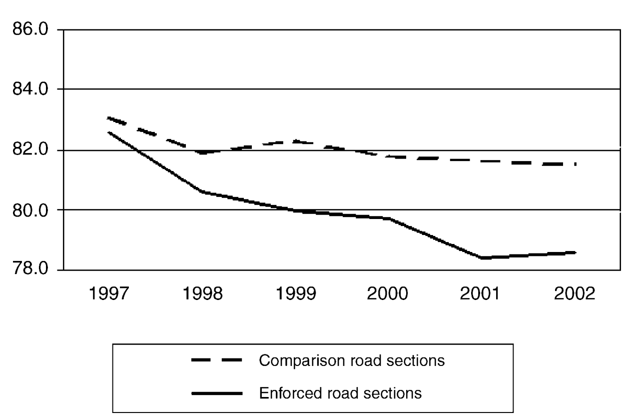

Fig. 1. Mean speeds on enforced and non-enforced 80 km/h roads in the period 1997–2002.

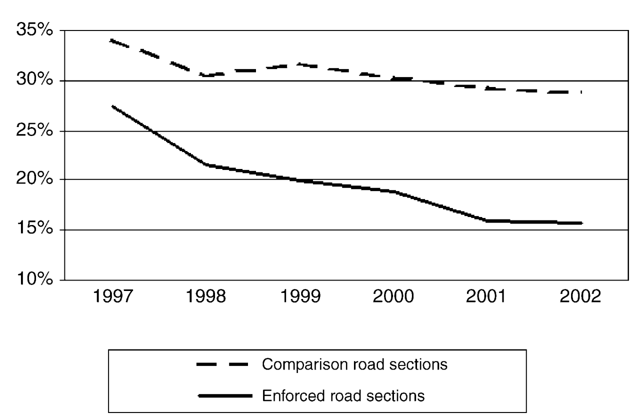

Fig. 2. Percentage of offenders on enforced and non-enforced 80 km/h roads in the period 1997–2002.

Table 6
Descriptive statistics of the speed data for the years 1997–2002

Indicator	Enforcement	1997		1998		1999		2000		2001		2002	
		M	S.D.	*M*	S.D.	*M*	S.D.	*M*	S.D.	*M*	S.D.	*M*	S.D.
Mean speed (km/h)	No	83.1	6.1	81.9	5.9	82.2	5.7	81.8	5.4	81.6	5.6	81.5	5.8
	Yes	82.6	3.7	80.6	4.7	80.0	4.5	79.7	4.3	78.4	4.5	78.6	4.3
% violators	No	34.1	14.7	30.4	13.5	31.7	12.7	30.3	12.1	29.2	12.5	28.7	4.2
	Yes	27.4	11.0	21.6	11.5	19.9	11.1	18.8	11.0	15.9	10.8	15.6	10.8

the start of the enforcement project in 1998, mean driving speed and percentage offenders have decreased from year to year (Figs. 1 and 2). For both mean speed and percentage of offenders, the largest effect sizes were found for the comparison between 1998 (start enforcement project) and 1997 (respectively $\eta^2 = 0.42$ and $\eta^2 = 0.53$), and for the comparison between 2001 (further increase of enforcement levels) and 2000 (respectively $\eta^2 = 0.38$ and $\eta^2 = 0.37$). Thus, the years of the start of the enforcement project and of the intensification of the speed enforcement were associated with the largest decrease in mean speed and percentage of offenders.

With regard to the mean speed, the interaction between time and enforcement was significant for the contrast between 2001 and 2000 ($F(1, 25) = 4.8$; $p = 0.037$; $\eta^2 = 0.16$). As can be seen in Table 6, this effect indicates that the further reduction in mean speed from 2000 to 2001 was larger on the enforced roads than on the comparison roads. The other interactions were not significant when contrasting the various years, nor were the interactions with regard to the percentage offenders. This indicates that the decrease between successive years was similar for enforced and comparison roads.

3.2. *Effects on road safety*

Table 7 shows the road safety developments at the enforced and comparison roads as well as the results of the odds ratio before/after comparison.

The odds-ratios were 0.79 (95% confidence interval, 0.66–0.95) for the number of injury accidents and also 0.79 (95% confidence interval, 0.63–0.99) for the number of serious casualties. This means that the best estimate is that there was an extra reduction of 21% for both the number of injury accidents and the number of serious casualties on the enforced roads. Expressed in absolute numbers, this indicates a saving of 50 injury accidents and 35 serious casualties over the 5-year period. Due to the number of observations, the confidence intervals around these estimates are large. However, with 95% certainty both odd ratios are below 1.0, hence there is a significant reduction in the number of accidents and casualties on the enforced roads.

The present evaluation study cannot completely rule out a number of other developments that may have influenced road safety on the enforcement roads. The most likely ones are the application of road engineering measures and the development of traffic flow over the years. With regard to

Table 7
Accidents en injuries per year, the before/after odds ratios (OR) and the 95% confidence interval (CI)

Year	Injury accidents (all severities) with at least one motor vehicle involved			Serious traffic casualties (fatalities + in-patients) of accidents with at least one motor vehicle		
	Enforcement group (per year)	Comparison group (per year)	OR* and 95% CI	Enforcement group (per year)	Comparison group (per year)	OR* and 95% CI
1990	50	494	0.79 (0.66–0.95)	42	314	0.79 (0.63–0.99)
1991	47	391		26	249	
1992	41	437		32	264	
1993	44	403		40	218	
1994	57	474		47	298	
1995	61	456		29	235	
1996	47	403		32	239	
1997	51	366		33	178	
Before	398	3424		281	1995	
1998	42	456		37	241	
1999	49	496		26	278	
2000	37	433		25	249	
2001	42	411		20	161	
2002	34	417		14	172	
After	204	2213		122	1101	

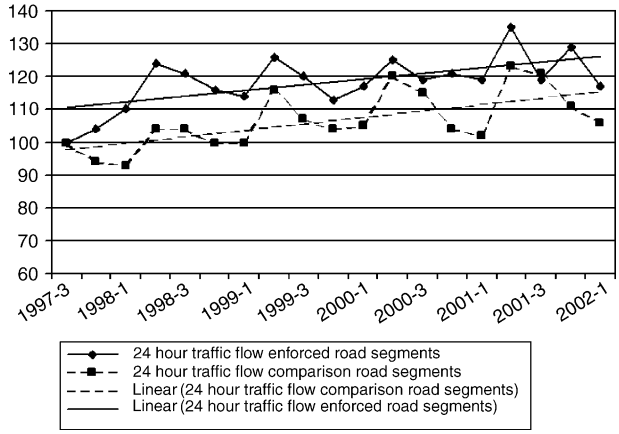

Fig. 3. Mean indexed (baseline: third quarter 1997 = 100) 24 h traffic flow and linear fit on 80 km/h road sections with and without speed enforcement per quarter in the period 1997–2002 (Source: loop measurements province of Friesland).

road engineering measures, there was no quantitative data to check whether more or less effective safety related measures had been taken at the enforced roads. With regard to traffic flow, the available speed data provided an indication of the development of the annual average 24 h traffic flow on both the enforced and comparison 80 km/h roads (Fig. 3).

For both groups of roads, there was an increasing trend in the average 24 h traffic flow over time. Based on linear fitted trend lines, the trend of increasing mobility was slightly smaller for the enforced roads ($y = 0.88x + 109.6$) than for the comparison roads ($y = 0.98x + 97.0$). Thus, it is unlikely that the favorable development of safety on the enforced roads can be explained by a different development of daily traffic on these roads. In addition, the average 24 h traffic flow of about 8200 vehicles at the enforced roads and 4300 vehicles on the comparison roads in 2002 was well below the capacity of this type of roads and, as such, is not expected to have a speed reducing effect.

4. Conclusions and discussion

The present study estimated the effects of 5 years of a regional speed enforcement program on rural roads in the Dutch province of Friesland. In this evaluation study, the effects on mean speed, the percentage of speed limit violators, the number of injury accidents, and the number of serious casualties were assessed by comparing the development on the roads that were subject to targeted speed enforcement with the development on similar roads without targeted enforcement.

Both the mean speed and the percentage of speed limit violators decreased during the targeted enforcement program. From a scientific point of view, the present study does not provide unequivocal statistical evidence that these effects are to be attributed to the speed enforcement activities. There may be two explanations for the lack of conclusive evidence on this point. Firstly, the limited number of roads in the enforcement and non-enforcement condition did not allow for a statistical test with a large discriminatory power. Secondly, there may have been a spillover effect of the enforcement from the enforced roads to the other, non-enforced roads. Also Keall et al. (2001) make reference to a possible spillover effect of mobile, inconspicuous speed enforcement. In the current study, the enforced roads and non-enforced roads were all located in one province and, hence, not very far apart. In addition, the enforcement program was supported by intensive general publicity in the regional media. As such, spillover effects are not unlikely. There was also some circumstantial evidence for this effect. Three roads in the comparison group showed particularly large decreases (more than 4 km/h) in mean speed between 1997 and 1998. On request, the regional road authority explained this phenomenon by the nearby presence of roads that belonged to the enforcement program. In addition, according to the road authority, on one of these three roads some speed checks may have taken place, however, not within the framework of the current project. All in all, the comparison roads were not a perfect control group. From this point of view, a comparison group of similar roads in another region of the Netherlands would have been better, but for practical and organizational reasons not feasible.

However, there are a number of (not statistically significant) indications that the speed enforcement affected vehicle speeds. During the enforcement program there was an almost continuous decrease in the mean speed and the percentage of offenders. In this period, the mean speed decreased with 4 km/h on the enforced roads and with 1.5 km/h on the non-enforced comparison roads. The percentage violators decreased with 12% points on the enforced and with 5% points on the comparison roads. Enforcement seems to be the most likely explanatory factor. In this period there were no other large-scale provincial or national road safety campaigns or programs focusing on speed or the dangers at rural roads. In addition, the largest decrease in mean speed and the number of offenders was at the start of the enforcement program in 1998 and after the further increase of the enforcement effort in 2001. At these times there were no sudden changes in traffic flow and, again, enforcement is the most likely explanation.

The number of road accidents and casualties decreased more at the enforced than at the comparison roads. Based on the available data, the best possible estimate of the traffic safety effect of the enforcement program is a 21% reduction of both serious casualties and injury accidents. However, this "best" estimate of the traffic safety effects should be viewed with some caution. Even though the estimate is based on a fairly long period, the absolute numbers of serious casualties and injury accidents are still small for statistical purposes. Hence, the actual effect may either be much larger or much smaller.

Another reason to be cautious about the estimated road safety effect is the potential influence of regression-to-the-mean. In this study, as in many other field studies of this kind, the roads were not assigned randomly to one of the two conditions. The roads that were part of the enforcement program were selected on the basis of their high number of injury accidents. In theory, it is possible that these high numbers reflected a temporary characteristic as a result of random fluctuations in accident numbers, and, hence, that the numbers would return to the overall mean in the period after. One way to correct for this regression-to-the-mean effect is the empirical Bayes method as proposed by Hauer (1997). In this method, the mean and variance of the expected accident numbers in a reference population is used to calculate the corrected estimate of the effect of an intervention. Unfortunately, in the current study it was not possible to find a good and sufficiently large reference population. The available traffic flow data indicate that the enforced roads had relatively large traffic volumes. The few remaining roads that were known to have comparable traffic volumes would have been far too small to make the required estimate of the expected accident numbers. Data of other roads, which would allow finding a larger and better reference population, was not available.

However, it is likely that the effect of regression-to-the-mean was limited in this study. The selection was based on accident data of a fairly long period of 5 years. In the 2 years preceding the selection period as well as the year following the selection period, the number of accidents was comparable to the average of the 5-year selection period (47, 50, 51 and 50 injury accidents, respectively). In general, the longer the selection period, the smaller the probability that high accident numbers are based on chance. Based on the work of Abbess et al. (1981), the "Road Safety Good Practice Guide" (DETR, 2001) states as a rule of thumb that with a 5-year selection period maximally 5–10% of the safety effect is to be attributed to regression-to-the-mean. Since in this study, the number of accidents was at the same level for a period of 8 years, it is unlikely that regression-to-the-mean played a crucial role. The fact that the enforcement roads seem to be rather exceptional with respect to traffic flow does, however, complicate generalizing the results. In fact, the effects of speed enforcement as found in this study specifically apply to rather busy and dangerous rural roads.

A last reason to be somewhat cautious about the size of the safety effect is that the influence of road engineering measures on the accident or injury risk cannot be completely ruled out. According to the qualitative information provided by the provincial authority, their role seems, however, limited (Goldenbeld et al., 2004). In the 5 year period of the current study, only a few engineering measures had been taken, both at the enforced roads and the non-enforced roads. In addition, some of these measures were taken or operative only at the end of the period under study, in 2001 or 2002. Moreover, many of these engineering measures (e.g. better lighting, roundabouts) could also have affected traffic safety on the nearby comparison roads, so that the influence would have spread out evenly over the enforced and the comparison roads. Traffic flow data showed that there is no reason to assume that migration of traffic from enforced to non-enforced roads explains the positive safety development at the enforced roads. It should also be noted that with an average 24 h flow of about 8200 vehicles on the enforced roads and 4300 vehicles on the non-enforced roads, the traffic flow is well below the point where it can be assumed to have a speed reducing effect. The roads in Friesland are almost never congested.

While keeping the above-mentioned reservations in mind, it is interesting to compare the safety estimate of this study with some estimates reported in other studies on speed enforcement at rural roads. Most of these studies concern fixed, visible speed cameras. For example in Norway, Elvik (1997) found a 20% reduction in the number of injury accidents in a study on the effects of fixed speed cameras at rural roads. A study in the UK (Hess, 2004) found a reduction of 21% in injury accident numbers when looking at the effects of fixed speed cameras at major rural roads within a 2 km distance at either side of the camera location. Closer to the camera sites the effects were substantially larger (ranging between 30 and 45%), but less comparable to the results of the present study that looked at road sections with an average length of about 4 km.

Even though the current study evaluated mobile cameras and not fixed cameras, the safety benefits are comparable. This is not the case if the results of a meta-analysis of studies in the period 1983–1996 by Elvik and Vaa (2004) are considered. Based on the analysis of three studies (six results), they reported an estimated effect of fixed speed cameras in rural areas of 'only' 4% (all accidents). Recent recalculations of these rural area data revealed, however, that the low estimate was incorrect and that the actual reduction in rural areas had to be 16% (Vaa, personal communication).

A number of studies specifically looked at the effects of mobile speed cameras in rural areas. In Canada, Chen et al. (2000) assessed the effects of mobile cameras at major rural roads and report a reduction of 25% in daytime speed related accidents, of 11% in serious daytime accident victims and of 17% in daytime accident fatalities. In New Zealand, Keall et al. (2002) found an additional reduction of mobile inconspicuous cameras (and an increase of around 20% of the speed camera operational hours) of 17% for injury accidents (not significant) and 31% for casualties in a 2-year period, as compared to a conspicuous camera program only, running elsewhere in New Zealand. In that study, the generalized effects of the extra hidden camera program (extending to the whole trial area containing (publicly) open rural roads, including roads with and without conspicuous camera operations) were estimated as an 11% reduction in accidents and a 19% reduction in casualties (both significant). In the UK, Gains et al. (2004) report on the results of an evaluation study of the British Safety Camera Program. With regard to mobile speed enforcement in rural areas they report a 15% reduction in the number of injury accidents. Even though each of the individual studies differ in several aspects, such as exact type of road, size of the enforcement area, type of accidents studied, length of the enforcement program, etc., the reported effects are generally very similar to the effects found in the current study.

The estimated 21% reduction of serious casualties and injury accidents in this study translates into the extra saving of about 35 serious casualties and 50 injury accidents in the 5-year period at the enforced roads. Due to the limited number of fatalities, a separate analysis on the reduction of fatalities was not possible. However, with an estimated reduction of 35 serious casualties, it is a very conservative estimate that at least two fatalities have been saved. Wesemann (2000) estimated the Dutch national economic costs of one traffic fatality (including the number of non-fatal victims and material damage coinciding with one fatality at a particular ratio) to be 6.6 million euros in 1997. After current correction for inflation, this estimate is 7.7 million euros. The total costs of the 5-year program were approximately 5 million euros. Therefore, even under the very conservative assumption that ‘only’ two fatalities (including associated non-fatal victims and non-material damage) were saved during these 5 years, the resulting benefit-cost ratio of the speed enforcement program of 3:1 would still be favorable.

Acknowledgments

We thank the following persons for providing data and background information for this study: J. Bruinsma (Team Traffic and Transport of the Province Friesland), K. Bunk (Public Prosecutor), M. van Essen (Bureau of Traffic Enforcement), S. van der Meulen (Regional Platform Road Safety Friesland), M. Popkema (Bureau of Traffic Enforcement), K. Post (police Friesland), S. Postuma (Bureau of Traffic Enforcement).

References

Abbess, C., Jarrett, D., Wright, C.C., 1981. Accidents at black spots: estimating the effectiveness of remedial treatment, with special reference to the ’regression-to-mean’ effect. Traffic Eng. Control 22 (10), 535–542.

Chen, G., Wilson, J., Meckle, W., Cooper, P., 2000. Evaluation of photo radar program in British Columbia. Accident Anal. Prev. 32, 517–526.

Christie, S.M., Lyons, R.A., Dunstan, F.D., Jones, S.J., 2003. Are mobile speed cameras effective? A controlled before and after study. Injury Prev. 9, 302–306.

Cohen, J., 1988. Statistical Power Analysis for the Behavioral Sciences, 2nd ed. Lawrence Erlbaum Associates, Hillsdale, NJ.

DETR, 2001. Road Safety Good Practice Guide. Department of the Environment, Transport and the Regions. The Stationary Office, London.

Durbin, J., Koopman, S.J., 2000. Time series analysis of non-Gaussian observations based on state space models from both classical and Bayesian perspectives. J. Royal Stat. Soc. B 62, 3–56.

Elvik, R., 1997. Effects on accidents of automatic speed enforcement in Norway. Transport. Res. Rec. 1595, 14–19.

Elvik, R., 2001. Cost-Benefit Analysis of Police Enforcement. Report ESCAPE project, Working Paper, VTT, Helsinki.

Elvik, R., Christensen, P., Amundsen, A., 2004. Speed and Road Accidents: An Evaluation of the POWER Model. Report 740/2004. Institute of Transport Economics (TØI), Oslo.

Elvik, R., Vaa, T., 2004. The Handbook of Road Safety Measures. Elsevier Science, Amsterdam, NY.

ETSC, 1995. Reducing Traffic Injuries Resulting from Excess and Inappropriate Speed. European Transport Safety Council, Brussels.

ETSC, 1999. Police enforcement strategies to reduce traffic casualties in Europe. European Transport Safety Council (ETSC), Brussels.

Evans, L., 2004. Traffic Safety. Science Serving Society, Bloomfield Hills, MI, USA.

Gains, A., Heydecker, B., Shrewsbury, J., Robertson, S., 2004. The National Safety Camera Programme. Three-Year Evaluation Report. PA Consulting Group, London.

Goldenbeld, Ch., Bijleveld, F.D., de Craen, S., Bos, N.M., 2004. Effectiviteit van snelheidstoezicht en bijbehorende publiciteit in Fryslân. (The effectiveness of speed enforcement and supporting publicity in the province of Friesland.) Report R-2003-27. SWOV Institute for Road Safety Research, Leidschendam, The Netherlands [in Dutch].

Hauer, E., 1997. Observational Before–After Studies in Road Safety. Elsevier Science, Oxford.

Hess, S., 2004. An analysis of the effects of speed limit enforcement cameras with differentiation by road type and catchment area. Transport. Res. Rec. 1865, 28–34.

Hess, S., Polak, J., 2003. Effects of speed limit enforcement cameras on accident rates. Transport. Res. Board 1830, 25–33.

Keall, M.D., Povey, L.J., Frith, W.J., 2001. The relative effectiveness of a hidden versus a visible speed camera programme. Accident Anal. Prev. 33, 277–284.

Keall, M.D., Povey, L.J., Frith, W.J., 2002. Further results from a trial comparing a hidden speed camera programme with visible camera operation. Accident Anal. Prev. 34, 773–777.

Kloeden, C.N., Ponte, G., McLean, A.J., 2001. Travelling Speed and the Risk of Crash Involvement on Rural Roads. Report No CR 204. Australian Transport Safety Bureau, Canberra, Australia.

Nilsson, G., 1982. The effects of speed limits on traffic crashes in Sweden. In: Proceedings of the International Symposium on the Effects of Speed Limits on Traffic Crashes and Fuel Consumption, Dublin. Organization for Economic Co-operation and Development (OECD), Paris.

Pilkington, P., Kinra, S., 2005. Effectiveness of speed cameras in preventing road traffic collisions and related casualties: systematic review. British Medical Journal, BMJonline, BMJ.com, doi:10.1136/bmj.38324.646574.AE.

van Schagen, I.N.L.G., Wegman, F.C.M., Roszbach, R., 2004. Veilige en geloofwaardige snelheidslimieten (Safe and credible speed limits). Report R-2004-12. SWOV Institute for Road Safety Research, Leidschendam, The Netherlands [in Dutch].

Stevens, J., 1996. Applied multivariate statistics for the social sciences, 3rd ed. Lawrence Erlbaum Associates Publishers, Mahwah, NJ.

Vaa, T., 1997. Increased police enforcement: effects on speed. Accident Anal. Prev. 29, 373–385.

Wesemann, P., 2000. Kosten van de verkeersonveiligheid in Nederland 1997 (Costs of traffic safety in the Netherlands 1997). Report D-2000-17. Institute for Road safety Research SWOV, Leidschendam, The Netherlands [in Dutch].

Zaal, D., 1994. Traffic law enforcement: a review of the literature. Report no. 53, Monash University, Accident Research Centre, Clayton, Victoria.

Zaidel, D., 2002. The impact of enforcement on accidents. Deliverable 3 (Work Package 2) of the ESCAPE project. Contract N: RO-98-RS.3047. http://www.vtt.fi/rte/projects/escape/escape_d3.pdf.